

**ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)**  
**ACADEX PRACTICE EXAMINATION**  
**A-LEVEL · PURE MATHEMATICS**

**PURE MATHEMATICS 6042/2**

**PAPER 2 November 2024**

**3 hours**

**PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED**

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2024 6042/2.

**Additional materials:** A calculator may be used unless a question forbids it.

**INSTRUCTIONS:** Answer all questions. Full marks are not given for answers only — show method.

- 1.** Find the stationary points of  $y = x^3 - 6x^2 + 9x + 1$  and determine their nature. **[10]**
- 2.** Find  $\int x e^x dx$  using integration by parts. **[10]**
- 3.** Solve  $dy/dx = 3y$  given that  $y = 2$  when  $x = 0$ . **[10]**
- 4.**  $x = 2t$ ,  $y = t^2$ . Find  $dy/dx$  in terms of  $t$ , and the equation of the tangent at  $t = 1$ . **[10]**
- 5.**  $A(1,0,2)$ ,  $B(3,4,2)$ . Find  $AB$  and  $|AB|$ . Show  $AB$  is perpendicular to  $(2, -1, 0)$  or not. **[10]**
- 6.** Show that  $f(x)=x^3 - x - 1$  has a root in  $[1, 2]$ . Take one Newton-Raphson step from  $x_0=1$ . **[8]**
- 7.** Solve  $|2x - 3| < 5$ . **[8]**
- 8.** Solve  $2 \sin \theta = 1$  for  $0 \leq \theta \leq 360^\circ$ . **[8]**
- 9.** A GP has first term 5 and common ratio  $1/2$ . Find  $S_\infty$  and the smallest  $n$  with  $S_n > 9.5$ . **[8]**
- 10.**  $x^2 + y^2 = 25$ . Find  $dy/dx$  at  $(3, 4)$ . **[8]**
- 11.** Find the area between  $y = 4x - x^2$  and the  $x$ -axis from  $x=0$  to  $x=4$ . **[8]**

- 12.** For small  $\theta$  in radians, write approximations for  $\sin \theta$ ,  $\cos \theta$ ,  $\tan \theta$ . Estimate  $\sin 0.1$ . **[8]**

**END OF QUESTION PAPER**

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).